MIL-G-48344 (PA)
28 May 1974
SUPERSESSION
(see 6.2)

MILITARY SPECIFICATION

GYPSUM, DEAD BURNED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers gypsum, dead burned, for use as an inert filler component in ammunition items.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FEDERAL

PPP-B-26 - Bag, Plastic, Polyethylene
UU-S-48 - Sacks, Shipping, Paper
RR-S-366 - Sieves, Standard for Testing Purposes
PPP-D-711 - Drum: Metal, Shipping, Steel, Lightweight (55 gallon)
PPP-D-723 - Drums, Fiber

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 6810

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation E300 - Industrial Chemicals, Recommended Practice for Sampling

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penn. 19103).

3. REQUIREMENTS

3.1 Material. The material shall be dead burned gypsum conforming to the following requirements:

3.1.1 Purity (as $CaSO_4$). The purity of the dead burned gypsum shall be 98 percent minimum (min) when determined as specified in 4.5.1.

3.1.2 Total Water Content. The total water content of the dead burned gypsum shall be 0.3 percent maximum (max) when determined as specified in 4.5.2.

3.1.3 Granulation (wet). Ninety seven (97.0) percent min of the dead burned gypsum shall pass through No. 325 U.S. Standard Sieve, when determined as specified in 4.5.3.

3.1.4 Foreign Material. The dead burned gypsum shall be free from extraneous foreign material such as metals, wood, minerals and other debris, when determined as specified in 4.5.4.

3.2 First Article Inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in

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the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of three (3) pounds of dead burned gypsum obtained by sampling as described in 4.4.2. The samples shall be obtained from a production batch which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be Performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.5 of this specification.

4.3.3 Rejection. See MIL-A-48078 (PA).

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. For the material covered by this specification, a lot shall consist of one or more batches of dead burned gypsum produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of dead burned gypsum that has been subjected to the same unit chemical or physical process.

4.4.2 Sampling. A representative sample of the material shall be selected from each batch. The tests specified in 4.5.1 through 4.5.4 shall be performed on the samples in accordance with MIL-STD-1235 CSP-1 Plan, Inspection Level II, AQL 6.5%. The size of the sample shall be approximately 500 grams and shall be withdrawn using ASTM Procedure E300 for solids. If any sample fails to meet any test requirement, the batch represented by the

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sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed shall be tested in accordance with the applicable methods given in paragraph 4.5. If any of these batches fail to meet any of the test requirements, that batch shall also be rejected. In addition, after any failure of a batch the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235. The classification shall be given in Table I.

TABLE I

Inspection/Requirement	Defect Classification
Purity (see 3.1.1)	Major B
Total Water Content (see 3.1.2)	Major B
Granulation (see 3.1.3)	Major B
Foreign Material (see 3.1.4)	Major B

4.4.3 Inspection Equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment shall be employed, where applicable for all tests and examinations specified in 4.5. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes.

4.5 Test Methods and Procedures. (see 6.3) The tests in 4.5.1 through 4.5.4 shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks.

4.5.1 Purity (as $CaSO_4$). Accurately weigh a portion of 0.25 to 0.30 grams (gm) of the sample and transfer

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to a 250 milliliter (ml) beaker containing 50 ml of 1:5 hydrochloric acid. Heat the contents of the beaker on a hot plate to boiling. Add 100 ml of boiling distilled water to the beaker and continue heating until the solution boils for five minutes. Filter the solution through a No. 1 Whatman filter paper, or equivalent, collecting the filtrate in a 250 ml beaker. Wash the residue on the filter paper with hot distilled water and collect the washings in the 250 ml beaker. Place the beaker and content on a hot plate. Bring the filtrate to a boil and slowly add 10 ml of a boiling solution of 10 percent barium chloride. Maintain the beaker and contents at low heat on a steam bath or a hot plate for one hour. Remove the beaker. Cool the solution at room temperature and filter the precipitate through a previously ignted and tared Selas procelain crucible of fine porosity. Then, wash the crucible and contents with distilled water until the washings show no turbidity when tested with silver nitrate solution. Dry the crucible and contents in an oven maintained at 105 $\pm$ 5°C for one half hour, ignite in a muffle furnace maintained at 650° + 50°C for twenty minutes. Remove from furnace, cool for one hour in a desiccator and weigh. Calculate the purity as follows:

$$\text{Purity as calcium sulfate percent} = \frac{58.33A}{W}$$

Where:

A = weight of ignited precipitate, gm
W = weight of sample, gm

4.5.2 Determination of total water. Transfer an accurately weighed 10 to 20 gm portion of the sample to a previously ignited and tared procelain crucible. Heat the crucible and contents in a muffle furnace maintained at 600 $\pm$ 50°C for two hours. Remove from muffle furnace, cool in a desiccator for approximately one hour and reweigh. Calculate the total water as follows:

$$\text{Total Water Content, percent} = \frac{(A - B) \times 100}{A}$$

Where:

A = weight of sample, gm
B = weight of ignited sample, gm

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4.5.3 Granulation (wet). Accurately weigh approximately 10 gm of the sample and place it on a tared 3 inch diameter No. 325 U.S. Standard Sieve. Immerse the sieve and contents in a flat bottomed dish or pan containing anhydrous isopropyl alcohol so that the alcohol covers the sample to the depth of one inch. Gently agitate the sieve until no material passes through the openings. Wash the sieve and contents with a stream of alcohol. Remove the sieve, containing the residue, from the alcohol and dry it on a steam bath until the odor of alcohol can no longer be detected. Then dry the sieve and residue in an oven at 100° to 105°C for 15 minutes. Caution: Completely remove the alcohol before placing in the oven, otherwise an explosive hazard exists. Weigh the sieve and contents and record the amount of material retained upon the sieve. Calculate the percentage of sample passing through the sieve.

4.5.4 Foreign Material. Spread the remainder of the sample on a flat surface and visually examine it for evidence of foreign materials. In addition, visually examine the residue remaining on the sieve after the granulation determination for evidence of foreign material.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A. The dead burned gypsum shall be packed in fiber drums conforming to Type III, Grade A requirements of PPP-D-723; metals drums conforming to PPP-D-711 for a Type IV drum or in paper sacks meeting the requirements of Specification UU-S-48 for a No. 14-5 ply or 14X-4 ply MB1 grade sack. Each fiber drum and metal drum shall be lined with a Type II, Grade A, Style 1, 2 or 3, PPP-B-26 plastic bag. Closure of the bag or drums shall conform to instructions contained in the applicable container specification. The gross weight of the fiber or metal drum shall not exceed 300 pounds. The net weight of the sacks shall not exceed 100 pounds.

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5.1.2 Level B. Packing requirements shall be the same as Level A.

5.1.3 Level C. The dead burned gypsum shall be packed in conformance with the latest uniform Freight Classification requirements.

5.2 Marking. Each drum or sack shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering Data - See MIL-A-48078 (PA).

6.2 Supersession data. This specification includes the requirements of Purchase Description PA-PD-722, dated 5 December 1955.

6.3 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P, Dover, New Jersey 07801. This description should include but not be limited to the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 6810-AA30

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